Fun as Moderate Divergence: Evaluating Experience-Driven PCG via RL

Ziqi Wang, *Member, IEEE*, Yuchen Li, *Student Member, IEEE*, Haocheng Du, *Student Member, IEEE*, Jialin Liu, *Senior Member, IEEE*, and Georgios N. Yannakakis, *Fellow, IEEE*

***Abstract*—The computational modeling of player experience is key to the generation of personalized game content. The notion of *fun*, as one of the most peculiar and core aspects of game experience, has often been modeled and quantified for the purpose of content generation with varying success. Recently, measures of a player's *fun* have been ad-hoc designed to model moderate levels of in-game divergence in platformer games, inspired by Koster's *theory of fun*. Such measures have shaped the reward functions of game content generative methods following the *experience-driven procedural content generation via reinforcement learning* (EDRL) paradigm in *Super Mario Bros* In this article, we present a comprehensive user study involving over 90 participants with a dual purpose: to evaluate the ad-hoc *fun* metrics introduced in the literature and test the effectiveness of the EDRL framework to generate personalized *fun Super Mario Bros* experiences in an online fashion. Our key findings suggest that moderate degrees of game level and gameplay divergence are highly consistent with the perceived notion of *fun* of our participants, cross-verifying the ad-hoc designed *fun* metrics. On the other hand, it appears that EDRL generators manage to match the preferred (i.e., *fun*) game experiences of each persona, only in part and for some players. Our findings suggest that the use of multifaceted in-game data, such as events and actions, will likely enable the modeling of more nuanced gameplay behaviors. In addition, the verification of player persona modeling and the enhancement of player engagement through dynamic experience modelling are suggested as potential future directions.**

***Index Terms*—Experience modeling, platformer games, procedural content generation (PCG), *Super Mario Bros (SMB)*, user study.**

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Ziqi Wang, Yuchen Li, and Jialin Liu are with the Research Institute of Trustworthy Autonomous System, Southern University of Science and Technology (SUSTech), Shenzhen 518055, China, and also with the Guangdong Provincial Key Laboratory of Brain-Inspired Intelligent Computation, Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: liujl@sustech.edu.cn).

Haocheng Du is with the Southern University of Science and Technology, Shenzhen 518055, China, and also with Tsinghua Shenzhen International Graduate School, Shenzhen 518055, China.

Georgios N. Yannakakis is with the Institute of Digital Games, University of Malta, 2080 Msida, Malta.

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I. Introduction

Offering personalized gameplay experiences through the procedural generation of game content remains a critical challenge in procedural content generation (PCG) research [1]. The challenge is many-fold but two core questions are particularly difficult to address. How do we measure player experience? How can we generate top-quality and appropriate content effectively and in real time? As a response to the above challenges, the recently introduced *experience-driven PCG via reinforcement learning* (EDRL) framework [2] offers a viable approach for the adaptive generation of player experiences in an online fashion. EDRL essentially combines the *experience-driven procedural content generation* (EDPCG) [3] and the *PCG via reinforcement learning* (PCGRL) [4] frameworks.

Inspired by Koster's *theory of fun* [5], a recent study [6] modeled *fun* as the moderate level of divergence across the creative facets [7] of level design and gameplay in platformer games. Such *fun* measures were used as reward functions that would train policies for the online generation of maximally-*fun Super Mario Bros* (SMB) levels via EDRL [2]. This article builds upon and significantly extends earlier work [6] across two main directions. First, we aim to validate the degree to which the proposed *fun* metrics are human-verifiable through a large-scale online user study of over 90 participants. Second, we test the effectiveness of personalized content generation across player types.[1] The key findings from the data analysis of the online user study presented in this article are as follows. First, the proposed *fun* measure that considers both game level and gameplay diversity is correlated highly with the human notion of *fun*, thereby validating its design and use for EDPCG in this game. Second, while showcasing some initial promising results, EDRL has room for further improvement in generating personalized *fun* experiences for specific player types and players. Third, future investigations are required on the use of multifaceted in-game data that would better capture human-like gameplay.

The rest of this article is organized as follows. Section II reviews related work in player experience modeling and human feedback. Section III presents the ad-hoc formulation of *fun* for SMB. Section IV introduces our developed platform for conducting user studies. Section V presents the protocol and the data collection process while Section VI discusses the key results

[1] All code, game levels, anonymous annotation data, and gameplay data used for the experiments of this paper can be found at the GitHub repository https://github.com/PneuC/Mario-Fun-Study.

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obtained, along with the core findings. Section VII provides a structured discussion and outlines promising future research directions. Finally, Section VIII concludes this article.

II. Related Work

Sections II-A and II-B review the literature on player experience modeling methods, and human subjective evaluations in games, respectively.

A. Player Experience Modeling

One of the core challenges of the EDPCG framework [3] is the computational modelling of player experience. A popular way to model the experience of a player is to construct computational functions that learn to map user and game data (e.g., game features, user behavior, facial expressions, or physiology) to annotated experiences of play. Shaker et al. [8] and [9] for instance, leverage neuroevolution to train a multilayer perceptron (MLP)-based player experience model (PEM) that predicts experience states, including *fun*, *frustration*, *challenge*, and *engagement*, of a specific player based on game content and player's statistical gameplay features. In a follow-up study, Shaker et al. [10] further improve the PEM by fusing head movement features with content and gameplay features. Blom et al. [12] track facial expressions using the INSIGHT SDK [11] to predict players' emotions in real time and dynamically adjust difficulty based on this information. Similarly, Teats et al. [13] employed preference learning to train an MLP-based PEM that predicts the subjective experience of *difficulty*, while the work of Joachims [14] used RankSVMs to predict *boredom* and *frustration* [15]. Zook et al. [16] make use of matrix factorization to predict the performance of a specific player, and the PEM is further used to guide the content generator to elicit some desired player performance curve. Finally approaching PEM from a designer's perspective, Liapis et al. [17] leveraged preference learning to evolve strategy maps interactively to a designer's taste.

While data-driven approaches can be used to construct efficient and accurate PEM, the shortage of player experience data can yield experience models that might not be able to generalize well. In such instances, data-free (e.g., top-down or ad-hoc design) approaches are more appealing and relevant. In that vein, some EDPCG studies attempt to model experience in an interactive manner. For instance, Hastings et al. [18] evolved game weapons in real-time based on the players' preference among an available set of weapons. Several other studies use *implicit* experience modeling methods by devising strategies to adjust the game's difficulty dynamically [19], [20], [21]. Plans and Morelli [22] devised a manually designed linear model to measure *excitement* based on gameplay features. They further develop a rule-based music generation engine to react to the prediction of excitement. Gravina et al. [23] model *surprise* using a computational model that measures "deviation from expectation," and they introduce search algorithms to maximize the surprise elicited by game content. Inspired by Saunders' *model of interest* [24], Merrick et al. [25] integrated a computational model of interest with shape grammars to generate combat quests for massively multiplayer online role-playing games. Shu et al. [2] introduced a divergence-based expression of *fun* following suggestions from Koster's *theory of fun* [5] and a *historical deviation* metric. They use the experience model to train an RL-based level generator [2]. Similarly to [2], Wang et al. [6] and [26] considered multiple facets and the players' episodic memory over time to train RL generators. In this work that builds upon and complements [6], we conduct a comprehensive user study to evaluate the degree to which the proposed model introduced in [6] is aligned with human notions of *fun*.

B. Human Feedback in Games

Human feedback plays an important role in content generation, for a number of reasons [1]. Importantly, human feedback obtained via user studies can be used to validate specific characteristics of the generated content and ensure that the content aligns with the intended design goals. Hervé and Salge [27], for example, compare PCG metrics with feedback received from human participants that judge *adaptability*, *functionality*, *narrative*, and *aesthetic* aspects of *Minecraft* settlement generation algorithms. In another recent study, a tower defence level generator is evaluated with regards to *gameplay*, *performance*, *difficulty*, and other aspects of the generated game through human feedback [28]. Mariño et al. [29] concluded that a well-designed user study is superior to well-designed computational metrics as the latter can only provide very limited to no information about visual aesthetics. Liu et al. [30] asked humans to score PCG-generated monsters in a double-blind trial and found out that their participants were unable to distinguish reliably between human-designed and PCG sequences of monsters.

Another core aim for the use of human feedback in PCG is to label the style, emotions, and experience of players while interacting with certain types of content (generated or not). This involves evaluating how players perceive and respond to the generated content in terms of, e.g., their emotional engagement, immersion, or enjoyment. Such evaluations can provide valuable insights into the overall player experience and help the algorithm to stylise the generated content accordingly [31], [32], [33]. Indicatively, Guzdial and Riedl [34] illustrated that *fun* experiences are personal and appear differently across typical Mario platformers, and their model successfully stylises the game content from gameplay videos; no measurement of *fun* or experience is proposed, however.

Alternatively, one may employ data-driven experience models to generate personalized content (see Section II-A). This approach involves asking human participants to label behavioral or affective aspects of game content based on their own subjective preferences. Such labels can then train surrogate models of preference and experience which can, in turn, drive the generation of personalised content [3], [12], [35], [36]. Indicatively, Schrum et al. [37] combined interactive evolution with latent variable evolution for generating more personalized game content. Shaker et al. [10] combined player gameplay data and visual cues from player's facial expression to predict labels of player experience; those models of experience were then used

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as personalized level generators that would, e.g., maximize a player's engagement.

Human feedback can be provided in various forms and across different means. Questionnaires define a common evaluation method across many game user research studies (e.g., [27], [28]) through which participants are normally required to visit a lab, complete an interactive session and then fill in a questionnaire in a post-experience manner. For wider participation and faster accessibility to experience labels, one may employ a custom-made survey platform, such as PAGAN [38]. A player experience annotation platform can either be employed *offline* (i.e., in the lab) or distributed *online* (i.e., in the wild) and thus employed to crowdsource human feedback [29], [34], [36], [38], [39]. The available annotation methods are as many as one's aims and the design needs of the experimental protocol. Ordinal (pairwise or ranking) annotation approaches are gaining in popularity due to their capacity to offer subjective labels of superior reliability and validity [12], [34], [39], [40], [41]. Based on a series of supporting studies about the benefits of ordinal annotation we employ both pairwise and ranking annotation schemes in this study. Moreover, we employ an online platform to solicit a representative and sufficient sample of human feedback regarding our experience metrics and their generated outcomes.

III. *Fun* as Moderate Divergence

According to Koster's *theory of fun*, *fun* in games relates to "learning in a context" [5], [42], i.e., *fun* occurs when players master new patterns. Based on this theoretical principle, it is essential for any level designer (human or automated) to be able to create divergent content that constantly introduces new gameplay patterns. The content experienced now by a player should be neither too different nor too similar to content that has been experienced in earlier phases of the game. Earlier work by Wang et al. [6] aimed to design and introduce a metric that formulates the characteristics of Koster's theory of *fun,* which was used as a platformer level evaluator. In particular, the metric we introduced in [6] is maximized when the diversity of a level and the possible gameplay patterns it enables are kept at a moderate degree. The rest of this section details the ad-hoc metrics introduced in [6] and cross-verified in this study.

A. Game Level Divergence

Game level divergence is formulated to measure the degree to which level segments differ in terms of structure and patterns. We adopt the *tile-pattern divergence* metric introduced in [43], which counts the number of all distinct local tile combinations (namely, tile patterns in a game level) and computes their frequency. We can then calculate the Kullback-Leibler divergence (KL-divergence) between any two tile-pattern frequency distributions of levels. While KL-divergence is insensitive to where patterns are located, it is sensitive to uncommon patterns. Using KL-divergence, however, requires an additional ϵ parameter to refine the frequency of uncommon tile patterns [43], as KL-divergence becomes infinite when considering frequencies of 0 value. Determining such a parameter can be challenging as it influences the sensibility of uncommon patterns. To address this limitation, the Jensen–Shannon (JS)-divergence [44] from the KL-divergence family is employed instead. JS-divergence, denoted as $D_L(\cdot,\cdot)$, is formulated as

$$D_L(A,B) = \sum_{x\in X}\left(p(x|A)\log\frac{p(x|A)}{p(x|A,B)} + p(x|B)\log\frac{p(x|B)}{p(x|A,B)}\right) \quad (1)$$

where A and B denote the two level segments to be compared, X denotes the set of all tile patterns that appear in A or B, $p(x|A)$ and $p(x|B)$ denote the frequency of a tile-pattern x that appears in A and B, respectively. $p(x|A,B) = \frac{p(x|A)+p(x|B)}{2}$ is the average frequency of tile pattern x appearing in both A and B. As in earlier studies [2], [6], the size of the tile pattern is empirically set to 2×2.

B. Gameplay Divergence

The play trace, i.e., the sequence of player coordinates over time during gameplay, is a popular way of representing a human gameplay demonstration [45], [46]. Similar to earlier studies in platformer games [6], [47], we use player traces to represent gameplay and measure the divergence between play traces through dynamic time warping (DTW) [48]. DTW is selected because it is widely used and can handle sequences of different lengths. Moreover, the use of DTW can eliminate overestimating the distance between sequences with shifting patterns. For instance, while the Hamming distance between the sequences ABCABC and BCABCA is 6, their DTW distance is 2. Formally, DTW attempts to capture gameplay divergence as follows. Let $\tau_i(A)$ and $\tau_j(B)$ be the ith coordinates of the play trace on the level segment A and the jth coordinates of the play trace on the level segment B, respectively. The distance measurement used for gameplay $D_G(\cdot,\cdot)$ is formulated as

$$D_G(A,B) = \min_{\mathbf{W}}\sum_{(i,j)\in\mathbf{W}} E\left(\tau_i(A),\tau_j(B)\right) \quad (2)$$

where $E(\cdot,\cdot)$ indicates the Euclidean distance between two vectors and $\mathbf{W} = [(i_1,j_1),\ldots,(i_t,j_t),\cdots]$ represents a warping path, which is a list of paired indices. A warping path satisfies three constraintsas follows:

1) it starts from $(1,1)$ and ends at (M,N), where M and N are the lengths of the traces on two level segments A and B, respectively;
2) it never goes backward or stop;
3) it covers every entry of the two sequences to be compared so that all elements of both play traces are considered in the alignment.

In our study, DTW will align the sequences according to the indices (corresponding to time ticks) of their coordinates $\tau_i(A)$ and $\tau_j(B)$. The use of both $\tau_i(A)$ and $\tau_j(B)$ coordinates allows DTW to capture more nuanced behaviors. Fig. 1 shows an illustrative example of computing DTW between two traces.

A play trace very much depends on a specific player and, given the dissimilarity of play, such traces may contain personalized behavior, which is often hard to predict. On the other hand, some

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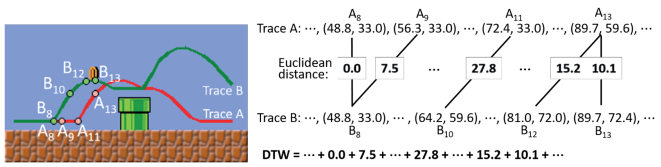


Fig. 1. Example of computing DTW between two traces.

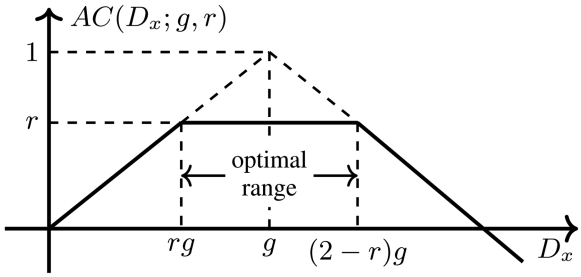


Fig. 2. Illustration of the A-clip function used for moderating the divergence of game levels and gameplay behavior.

games typically have tight rules and goals, thereby regulating and often limiting the options for the player. Regardless, in this work, we assume the in-game agent playing the generated level is tightly coupled with the different ways a player would behave and experience a game level. We thus borrow three dissimilar A*-based agents from [49] to simulate gameplay in different ways that are outlined here. The *Runner* agent always tries to complete levels as fast as possible, while the *Killer* and *Collector* agents, respectively, prefer to kill as many enemies and collect as many coins as possible. Although simulated play traces from procedural personas and play traces from human players are expected to differ, the simulated play trace can be assumed to capture a prototypical gameplay behavior with an underlying utility as in earlier works [6], [32].

C. Moderating Divergence

To capture the notion of "moderate divergence" [2], we first define an *A-clip function* [6] as follows:

$$\mathrm{AC}(D_x; g, r) = \min\left\{r, 1 - \frac{|D_x - g|}{g}\right\} \tag{3}$$

where D_x can be either D_L or D_G, or even other divergence metrics, g and $r \in (0, 1]$ are parameters describing the target value and the clipping rate of this function, respectively. When the input D_x lies within the optimal range of $[rg, (2-r)g]$, the function returns its maximum value of r (see Fig. 2).

The A-clip function is used to evaluate the divergence values between a newly generated segment and a number of segments played in the past separately. In the A-clip function, the clipping rate r controls the span of the optimal range and also the upper bound.

When designing a function of player experience we assume that the player's *memory* of experiencing earlier level segments decays over time due to episodic memory. Based on this assumption, earlier game segments are assigned a lower clipping rate, allowing for a wider optimal range and a smaller maximum value. Formally, the work in [6] defines the moderate degree of divergence in play (i.e., "*fun*" as in [5]) as the following *fun* metric denoted as f_x

$$f_x(S_i) = \frac{\sum_{k=1}^{n} \mathrm{AC}(D_x(S_i, S_{i-k}); g_x, r_k)}{\sum_{k=1}^{n} r_k} \tag{4}$$

where n determines how many previous segments will be considered, i.e., assuming a human player typically maintains a memory of the n earlier segments. $r_k = 1 - \frac{k}{n+1}$ and x indicates the facet being considered. By substituting D_x with the game level divergence measurement D_L [see (1)] and gameplay divergence measurement D_G [see (2)] in (4), respectively, we formulate the level-based *fun* metric f_L and the gameplay-based *fun* metric f_G. Parameter g_x determines the centre of moderating divergence (see Fig. 2). The value of r_k decreases as k goes larger, which means that the optimal range of moderating divergence between S_i and S_{i-k} keeps slacking. It also indicates a decline in importance since the maximal value of the A-clip function equals r. The g_x value is a fixed parameter, set to be consistent with respect to k. Specifically, g_L and g_G are, respectively, set to 0.14 and 0.3, as suggested in [6]; n is arbitrarily set to 5. Each generated level is split into fixed-size segments of 16-tile width. Readers are referred to [6] and our code[1] for further details concerning (4).

IV. Mario Survey Platform

This section introduces the online SMB annotation platform we devise and implement for conducting human studies with our EDRL generators.

The online annotation platform facilitates a complete user study protocol, including game-playing and annotation. Participants can access the platform openly through the internet,[2] making it accessible to a wide range of potential participants. Upon accessing the platform, participants are guided through the entire experimental protocol by the platform itself. They are directed to play games and annotate their experiences without any additional guidance or intervention. The platform is designed and validated to be user-friendly and intuitive, ensuring (to the best degree possible) that participants can easily navigate and complete the study tasks without any confusion or difficulty. No participant in our pilot user studies reported any difficulties in terms of navigation and task completion.

The platform provides three main functionalities: *questionnaire*, *gameplay,* and *annotation* as shown in Fig. 3 and outlined in the rest of this section.

A. Questionnaire

Participants were required to complete the questionnaire before proceeding further to the next steps of the user study. In particular, we collect demographic information regarding their age, gender, gameplay persona, and the frequency of playing SMB, platformer games, and games in general.

[2]The platform is available at: www.mariogametest.top

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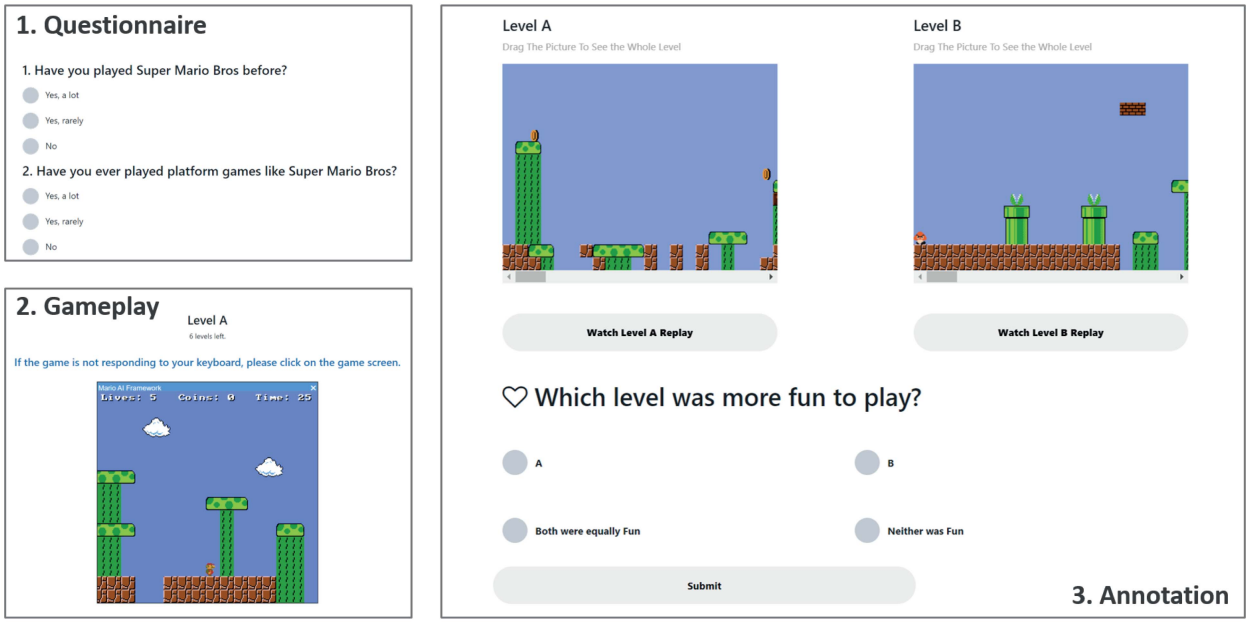


Fig. 3. Overview of the online SMB annotation platform.

B. Gameplay

The gameplay functionality is implemented using a widely-used open-source version of SMB [50] written in `Java`. The game is embedded into the platform through `Cheerpj` [51], allowing participants to play SMB levels directly on web browsers, similar to desktop applications. We extend the engine with new APIs to encode gameplay actions as binary strings and save them into a single file, which can be loaded into the extended game engine to reproduce the entire gameplay, providing records of essential data for analysis. The platform can thus offer video replays for participants to view and annotate via a *stimulated recall* protocol [52].

C. Annotation

As already mentioned, the platform supports the *annotation* of game content, through which participants are asked to evaluate the experience of the levels they just played. We adopt the ordinal annotation approach to assess the value of player experience given its benefits to yield more reliable affect labels [41], [53], [54]. The current implementation of the platform offers two types of ordinal annotation: *pairwise* and *ranking*. The pairwise annotation protocol adopts the four-alternative forced choice method employed in many studies of player experience modelling [1] and asks participants to indicate their *fun* preference between a pair of levels (i.e., "*Which level was more fun to play?*"). To account for cases where the levels are perceived as equally fun or equally nonenjoyable, the question offers two additional possible answers: "*Both were equally fun*" and "*Neither was fun.*" The ranking annotation scheme is inspired by the work in [17] and requests participants to rank multiple levels in terms of *fun* through a dragging list. This method allows participants to provide a more nuanced evaluation of the levels they played.

To ensure that participants accurately recall the details of each level and their playing experience, the annotation page provides images and videos of the levels, accessible through the *replay* function described in Section IV-B. Participants can review the entire level by dragging the scroll bar and also replay specific levels by clicking the replay button on the annotation page. These features are aimed to assist participants (i.e., as triggers to their episodic memory) to provide labels that are as close as possible to the actual gameplay experience.

V. User Study

A user study protocol is designed with the following two research questions in mind.

1) To which degree are the proposed *fun* metrics correlate with human annotated notions of *fun*?
2) Can these metrics enable EDRL to maximise the *fun* for specific player types thereby personalizing player experience?

To address the above research questions, we designed two sets of experiments, referred to as *Experiment-I* and *Experiment-II*. Briefly, in Experiment-I, participants play two levels with dissimilar *fun* values in terms of f_L and f_G, respectively, and then label which one is more *fun*. In Experiment-II, participants play a set of three levels, each generated by a different EDRL generator trained with playing agents of different playing styles, and then rank the levels. Section V-A describes the process of generating and selecting game levels used in the experiments. Section V-B details the experimental protocol. Section V-C summarizes the information of participants and the data cleaning process. The user study was performed through the platform described earlier in Section IV.

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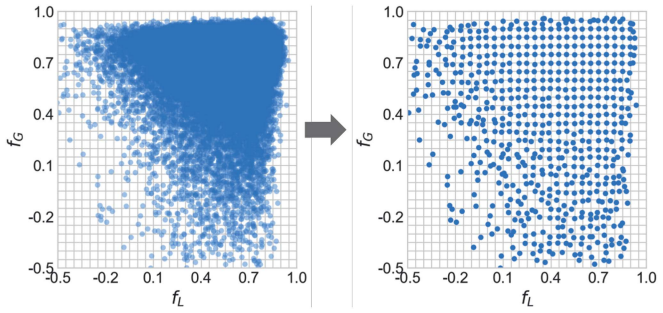


Fig. 4. Selection process of levels for Experiment-I. Scatter plots show the distribution of levels in the f_x metric space (f_{L} versus f_{G}). Initially, 50 000 playable levels are generated (left) and 633 are ultimately selected so that the f_{L}-f_{G} space is covered uniformly (right).

A. Game Levels

Sections V-A1 and V-A2 detail the two sets of levels for Experiment-I and Experiment-II, respectively.

1) Levels of Experiment-I: To verify if the proposed *fun* metrics are correlated with human annotated notions of *fun*, we select generated levels of variant *fun* values as follows. Inspired by the authors in [2] and [55], we generate all game levels used in our experiments via a generative adversarial network (GAN). Specifically, the GAN is trained to map a low-dimensional continuous vector from a predefined latent space to a one-hot tensor, which represents a level segment exclusively. We employ the self-attention GAN architecture [56], which has shown promising performance in generative tasks [57]. Each generated game level is composed of 11 level segments sampled from the latent space of the trained GAN. These segments are first concatenated into a complete game level, and then their playability is checked using the *Runner* agent from [49]. If the agent can complete the level, the latter is added to the set of playable levels. The procedure is repeated until a total of 50 000 levels are added to the playable set.

Next, we select representative levels from the playable level set. A 2-D map is initialized with 30×30 cells, representing a 30×30 uniform partition of the metric space with $f_{\mathrm{L}} \in [-0.5, 1.0]$ and $f_{\mathrm{G}} \in [-0.5, 1.0]$. Then, all playable levels are placed into the cells on the 2-D map according to their f_{L} and f_{G} values (cf., Section III). Finally, the level with the highest $f_{\mathrm{L}} + f_{\mathrm{G}}$ value in every cell is preserved, and the others are filtered out. The same selection criterion is applied to every cell, thereby yielding a selection process that is as a uniformly distributed as possible.

This process results in a total of 633 levels, distributed uniformly within the 2-D metric space. Fig. 4 illustrates the distribution of all the 50 000 levels and the 633 representative levels as scatter plots of f_{L} and f_{G}.

2) Levels of Experiment-II: For Experiment-II, we utilize the EDRL framework [2] to train three distinct level segment generators, namely R, K, and C corresponding, respectively, to playstyle *Runner*, *Killer*, and *Collector* agents designed by Zhang et al. [49]. This framework conceptualizes online EDPCG as a Markov decision process, where the main objective is to generate engaging game levels. The EDRL framework uses a GAN as a decoder. This GAN transforms low-dimensional latent vectors into one-hot representations of level segments. In this setup, an RL designer observes the latent vectors of the five most recent level segments to decide on the next segment. The chosen action is a latent vector that the GAN decodes into a segment. The reward in this framework is a combination of *fun* metrics (denoted as f_L and f_G) and *playability* rewards (denoted as P). Specifically, playability is marked as -1 for any segment that the *Runner* agent cannot complete within the allowed decision time, and 0 otherwise. This system effectively penalizes the policy for creating segments that are not playable.

TABLE I
AVERAGE AND STANDARD DEVIATION OF f_L AND f_G VALUES OF THE LEVELS GENERATED BY THREE DIFFERENT EDRL GENERATORS

Generator	f_{L}	f_{G}^R	f_{G}^K	f_{G}^C	P
Runner	**.802±.040**	**.837±.061**	*.724±.250*	*.706±.184*	**-.005±.040**
Killer	*.783±.041*	.793±.089	**.857±.110**	.752±.588	-.005±.041
Collector	.786±.036	*.779±.105*	.854±.128	**.780±.289**	*-.016±.036*

Gameplay-based f_G values are evaluated across all three playing agents independently. The playing agent used is denoted by the superscript of f_G.

The training of the GAN follows the approach introduced in [56], employing a self-attention GAN architecture and hinge adversarial loss. The training set is composed of segments of size 16×16 obtained by sliding a window with stride of one onto all the 15 human-authored levels from the Mario AI framework [50]. To develop the level generators, we trained three different policies by each type of agent along with the same reward function $f_L + f_G + P$. Each generator is paired with a distinct agent that simulates different playing styles, where R is trained with the *Runner* agent, K with the *Killer* agent, and C with the *Collector* agent. These agents help us to calculate f_G by simulating play traces and are designed to reflect the perceived fun of their corresponding personas. The agents have been trained using the episodic generative soft actor–critic introduced in [6], over a span of one million time steps. We parallelized the environment using 50 synchronous processes and apply the update method of the soft actor–critic using a randomly sampled batch of size 256 every time two transitions are collected. Target entropy is set to -20 as recommended in [58] and smoothing coefficient τ is set to 0.002. Discount rate γ is 0.8 as suggested in [59].

Each EDRL generator creates 100 game levels, where each level is composed of 11 segments. The generation process starts from 100 randomly chosen latent vectors that have been validated for playability. The f_x values of the three groups of levels are evaluated as shown in Table I.[3] These evaluations show that the generators are capable of producing levels that are not only playable but also score high on the f_x metrics. However, negative values of P indicate that there were some levels the *Runner* agent failed to complete within the allotted time. We

[3]From now on, the generators trained with *Runner*, *Killer* and *Collector* playing agents (and levels generated by those generators) are referred to as the R, K and C generators (and levels), respectively.

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manually reviewed the playability of these levels and excluded the unplayable ones from the results presented in Experiment-II.

B. Experimental Protocol

Participants are first presented with a welcome screen that explains the purpose of the study and details the tasks. They then proceed to complete a demographics questionnaire that gathers information about their *gender*, *age*, and their preferred *play style* (referred to as *persona*) in SMB. The *persona* question offers participants four options: "*Runner: trying to complete a level as fast as possible by running and jumping,*" "*Killer: prefer killing all the monsters in a level,*" "*Collector: attempt to collect all the coins in a level,*" "*None of the above.*" Participants select the option that best represents their preferred play style. Based on their selection, they are identified as *Runner*, *Killer*, *Collector*, or *Not applicable*, respectively. This classification of player personas is adopted according to the designed agents' persona, as mentioned in [49]. Soon after they fill in the demographics questionnaire, participants are requested to complete the two dissimilar experimental sessions (a total of seven levels) by playing SMB games and annotating their experiences on the levels they just played.

In the first experimental session (namely, Experiment-I), participants play two pairs of levels (in a total of 4 levels) randomly selected from the first set of generated levels (cf., Section V-A1). After playing both levels (A and B) in a pair, participants are presented with two videos of their gameplay allowing them to review the two levels and their corresponding gameplay as much as needed. The participants are then asked "*Which level was more fun to play?*" They may select one out of four choices (4-alternative forced-choice protocol): "*A,*" "*B,*" "*Both were equally fun,*" and "*Neither was fun.*"

The second experimental session (namely, Experiment-II) follows a similar structure. Specifically, participants play a set of three levels, each generated by a different EDRL generator as described in Section V-A2. After playing the levels, participants rank them based on their perception of *fun* using a drag-and-drop list (as in [17]). The order in which the levels are played is shuffled and participants are not informed about the ways levels are generated. Upon completion of the annotation task, participants have the option to play and annotate another set of levels. This design allows for multiple rounds of data collection and helps solicit more human feedback.

C. Data Collection and Cleaning

Our participants were invited through social media campaigns and the 2023 IEEE Conference on Games. In total, 193 participants accessed the website and filled in the questionnaire. Their collected data consists of five types including demographic questionnaire responses, Experiment-I annotation labels, Experiment-II annotation labels, gameplay statistics, and data for replaying the level. Once data are collected, we go through a thorough data cleaning process to ensure completeness, data integrity, validity, and to reduce noise. Specifically, any participant who failed to complete the first two segments (i.e., the minimum number of segments that allows f_x value computation) of a level is removed from the dataset. Moreover, questionnaire responses that have no associated annotation labels are removed. This ensures that only the participants who contributed with at least one annotation task are considered.

VI. Results and Analysis

This section reports the participants' responses to demographic questionnaires, and the analysis of findings across the two experiments designed.

A. Participants and Their Demographics

After the data cleaning process described in Section V-C, we are left with 93 filled-in questionnaires, and 145 and 97 annotation labels for Experiment-I and Experiment-II, respectively, that are considered valid for further analysis. The majority of our participants (76 out of 93) finished the survey entirely (i.e., both Experiment-I and Experiment-II). Among the 93 participants who offered valid data, 20 identified as female, 64 as male, and 9 as nonbinary or preferred not to say. The age of the participants ranges between 17 and 47 years. Only 10 out of our 93 participants had never played SMB before, and 5 had not played SMB but had experience with other platformer games similar to SMB. On the other hand, 33 participants had extensive experience playing SMB. In addition, 26 participants reported playing games for more than 10 h a week, while 15 indicated that they rarely played games, and 2 stated that they never played games regularly.

B. Experiment-I

Inspired by earlier studies on skill stratification [60], before any further data analysis is conducted, we first divide participants into two groups based on their in-game performance. Performance is determined by the average completion rate (CR) across all the levels a participant played. The upper half of the participants with the highest average CR are grouped as *expert*, while the remaining participants are grouped as *nonexpert*. The CR is calculated as the proportion of the level completed by players compared to the total level length. While the number of annotations in these two groups is supposedly similar, expert players contribute more annotations because nonexpert players fail to complete the first two segments far more frequently. Consequently, some of their annotations are removed during the data cleaning process before analysis.

For each pair of levels A and B used in A/B test, the difference of their f_x values is calculated as

$$\Delta f_x = \frac{1}{n'}\sum_{i=1}^{n'} f_x\left(S_i^{\mathrm{A}}\right) - \frac{1}{m'}\sum_{i=1}^{m'} f_x\left(S_i^{\mathrm{B}}\right) \tag{5}$$

where n' and m' are the number of segments players have played in levels A and B, respectively.

To assess the degree of agreement between f_x values and the human perception of *fun*, we define *agreement* when the level preferred by a player has a larger f_x value compared to the nonpreferred level, and *disagreement* if otherwise. Figs. 5 and 6 provide visual representations of the annotation labels

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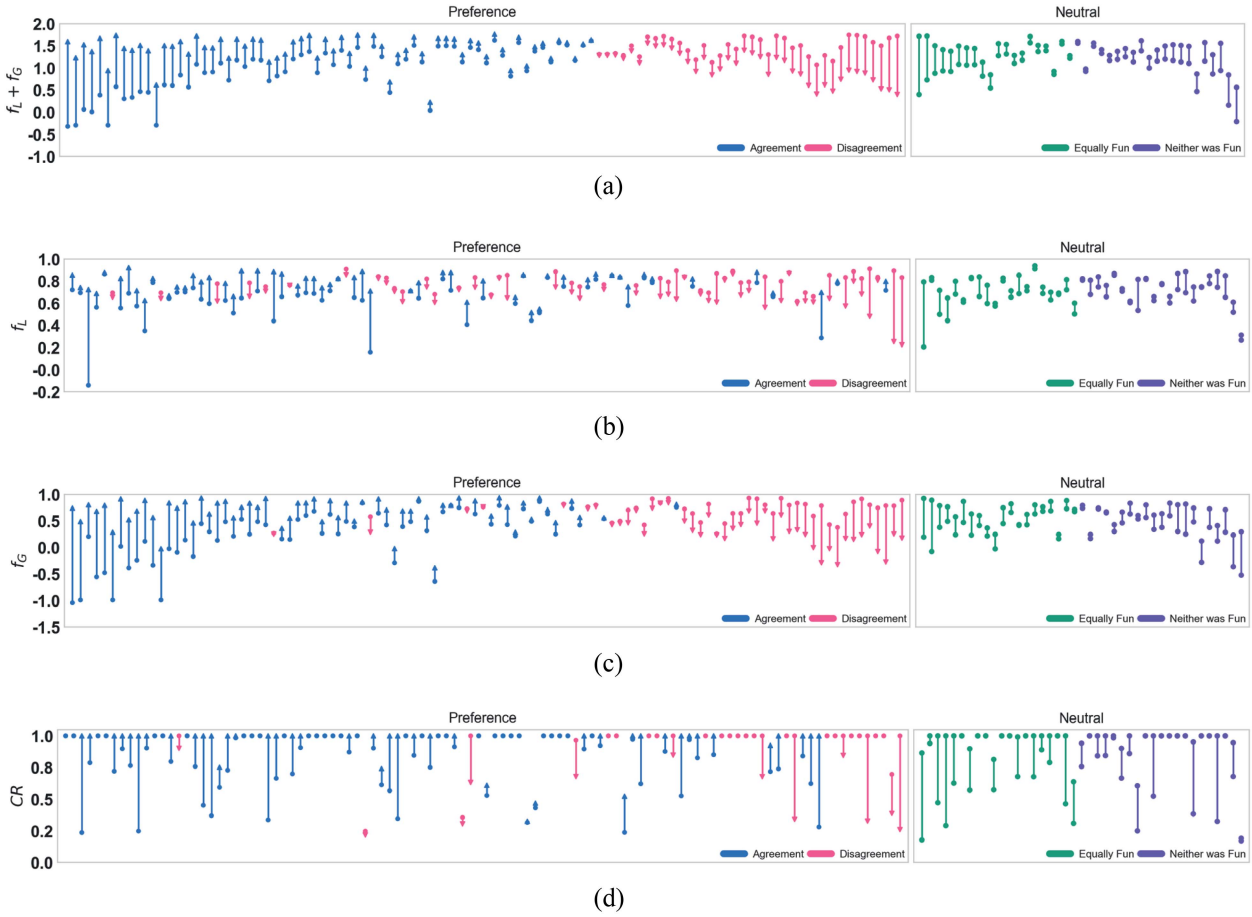


Fig. 5. Visualization of human feedback (*fun* preferences) for Experiment-I. Each vertical bar represents a *fun* preference of a pair of levels. The two endpoints of each vertical bar indicate the f_x values of the two levels of the pair. The subplots titled "Preference" show all clear preferences (i.e., "A" and "B" responses), in which the arrow points to the preferred level of the pair. The subplots titled "Neutral" show all pairs of levels that were annotated with *equally fun* or *neither was fun* labels. For visualization proposes, we sort the pairs by the value difference of $f_L + f_G$. All other graphs follow the same order on the x-axis. (a) *Fun* preferences compared against combined game level and gameplay *fun* measure ($f_L + f_G$). (b) *Fun* preferences compared against combined game level and gameplay *fun* measure f_L. (c) *Fun* preferences compared against gameplay *fun* measure f_G. (d) CR differences.

with corresponding *fun* values obtained from Experiment-I, illustrating their relationship for all groups and for expert and nonexpert groups. Particularly, for each pair of levels played, we compute the level-based f_x value f_L, the gameplay-based f_x value f_G, and their combination $f_L + f_G$ for all players (expert and nonexpert players). The figures also illustrate the corresponding preferences of each game level pair as annotated by its players. We take into consideration the assumption that players may prefer levels they are more likely to complete. For that purpose, we also illustrate the game CR to triangulate the relationship between *fun* metrics, player preferences, and in-game skill. Figs. 5(a) and 6(a) locate *agreement* pairs to the left, and *disagreement* pairs to the right of the graph. The order within each group is descendant toward the centre of the graph and it is based on the $|\Delta f_x|$ value of the pair. For ease of comparison all other subfigures showcase results based on this particular order.

Based on the data presented in Fig. 5(b), we can observe that the f_L values exhibit a weak correlation with the annotated preferences as the numbers of disagreements and agreements are similar (i.e., 29 agreements against 28 disagreements). Fig. 5(a) and (c), however, illustrate a different image as agreements appear to be far more than disagreements, suggesting a positive and high correlation between the values of f_G and $f_L + f_G$ and players' perception of *fun*. Moreover, we observe a key difference in the number of labels provided by experts versus nonexpert players in Fig. 6. Nonexpert players tend to have more neutral preferences compared to expert players. This suggests that skilled players can define their *fun* preferences between two levels more easily in the game.

To evaluate the degree of human verification of the proposed metrics, we calculate the correlation between the metrics and the players' perception of *fun* using Kendall's τ coefficient [61]. Specifically, τ is defined as $\frac{n_+ - n_-}{n_+ + n_-}$, where n_+ and n_- are the number of concordant (agreement) pairs and the number of discordant (disagreement) pairs, respectively. The values of τ lie in $[-1, +1]$, where -1 and $+1$ indicate a complete disagreement and agreement, respectively. Table II shows the results of Kendall's τ coefficients for each metric. The positive τ values for all metrics and all groups indicate that all the metrics are positively correlated to players' perceived notion of *fun*. Specifically, the τ values for $f_G + f_L$ show a significant correlation of the metric with the perceived notion of *fun* within both groups of experts and nonexperts and among all players.

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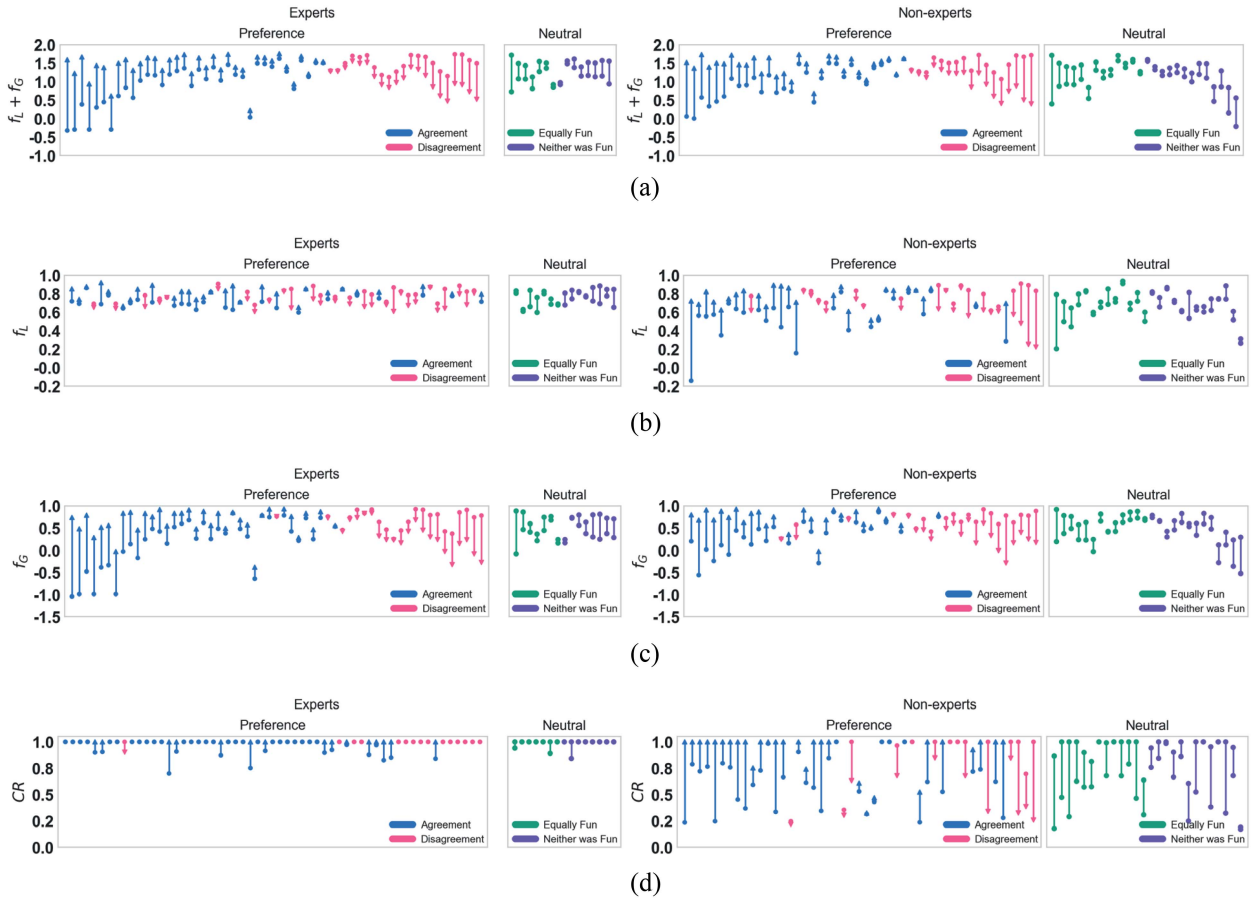


Fig. 6. Visualization of human feedback (*fun* preferences) for Experiment-I. Each vertical bar represents a *fun* preference of a pair of levels. The two endpoints of each vertical bar indicate the f_x values of the two levels of the pair. The subplots titled "Preference" show all clear preferences (i.e., "A" and "B" responses), in which the arrow points to the preferred level of the pair. The subplots titled "Neutral" show all pairs of levels that were annotated with *equally fun* or *neither was fun* labels. For visualization proposes, we sort the pairs by the value difference of $f_L + f_G$. All other graphs follow the same order on the *x*-axis. (a) *Fun* preferences compared against combined game level and gameplay *fun* measure $f_L + f_G$. (b) *Fun* preferences compared against gameplay *fun* measure f_L. (c) *Fun* preferences compared against combined game level and gameplay *fun* measure (f_G). (d) CR differences.

TABLE II
KENDALL'S τ COEFFICIENTS BETWEEN THE *FUN* METRICS AND PARTICIPANTS' PREFERENCES

Metric	Expert			Non-expert			All		
	n_+	n_-	τ	n_+	n_-	τ	n_+	n_-	τ
f_L	29	28	0.018	27	20	0.149	56	48	0.077
f_G	35	22	0.228*	26	21	0.106	61	43	0.173**
$f_L + f_G$	36	21	0.263**	30	17	0.277**	66	38	0.269**

n_+ and n_- denote the number of agreements and the number of disagreements, respectively. * indicates significance (p-value < 0:05), while ** indicates high significance (p-value < 0:01).

The τ values for f_L, however, do not yield any statistical significance. Even though the game level-based *fun* metric does not seem to correlate highly with annotated *fun* when combined with the gameplay-based metric it yields high and significant correlation values with human *fun* preferences. Ultimately, the *fun* metric designed and used in [6] yields the highest correlation with *fun* preferences compared to the other two metrics independently.

C. Experiment-II

We consider two ways of clustering players into groups for the analysis of data obtained from Experiment-II. The first *subjective* approach relies on self-reported player style (cf., Section VI-C1) whereas the second *objective* approach clusters players based on their in-game behavior (cf., Section VI-C2).

1) Group Players Subjectively via Self-Reports: We categorize participants into three different play style groups (*Runner*, *Killer*, and *Collector*) based on their responses to the questionnaire. We hypothesise that each group will prefer the levels generated by the corresponding generator. Therefore, we plot the pie chart (cf., Fig. 7) for each persona and highlight the agreement cases in navy color where participants rank the level generated by the corresponding generator as their first choice. Notably, the majority of *Runner* players (i.e., 79%) did not prefer the levels generated by the R generator. This suggests a potential mismatch between the levels tailored to their play style and players' actual preferences. Similarly, 64% of *Killer* players preferred levels generated by other generators over the levels from the K generator. The discrepancy is less apparent

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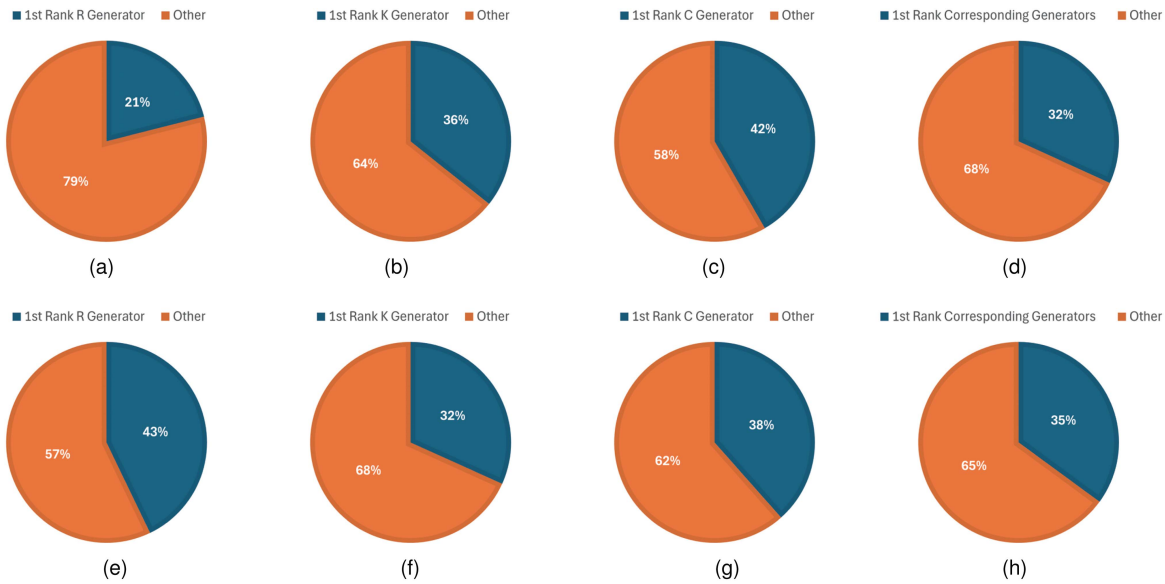


Fig. 7. Proportion of participants that rank the generated level of their corresponding persona as the most *fun*. Participants are categorized according to their player types identified by the questionnaire responses (top row) and by their in-game behaviors (bottom row). The number of players being identified as a certain type is given between parentheses. Note that nine participants identified themselves as "none of those three personas" when answering the questionnaire. (a) Self-reported: Runner (38). (b) Self-reported: Killer (14). (c) Self-reported: Collector (36). (d) Overview (88). (e) Behavior: Runner (21). (f) Behavior: Killer (63). (g) Behavior: Collector (13). (h) Overview (97).

with the *Collector* players as nearly half (i.e., 42%) of those players preferred levels generated by the C generator. Across all player types, 71% did not rank the level from the corresponding generator as their top choice, and only 29% showed a preference for levels aligned with their identified persona.

The data suggests that the levels generated by personalised generators did not strongly resonate with the respective persona. A significant portion of players across all types preferred levels generated by other types of generators. This may reflect a need for a deeper understanding of what each player type truly values in game levels, or it might suggest that agents do not capture the players' nuanced behavior. Section VII provides more in-depth discussions of those results and potential future directions.

2) Group Players Objectively via In-Game Behaviour: Assuming that self-reported player style might deviate from the player's actual in-game behavior, we cluster participants based on their in-game behaviors and repeat the analysis. We first contrast the player's trace against all play traces of the three persona agents on each level. DTW is used for measuring the pairwise similarity of the traces. The persona with the smallest DTW distance value is assigned as the player's persona.

The results shown in Fig. 7 indicate an improved alignment between players' in-game behavior-based personas and the game levels generated for them compared to the alignment with self-reported personas. Contrary to expectations and earlier analyses, 43% of *Runner* players prefer R generated levels, while 57% favour levels from other generators. Similarly, the *Collector* players show a somewhat looser alignment, with 38% of them preferring C generator levels However, the majority of *Killer* players (i.e., 68%) do not rank K generator levels as their top preference. Overall, 35% of all players indicate their highest preference for levels from the generator corresponding to their assigned personas.

TABLE III
NORMALISED AVERAGE RANKS WITHIN EACH GROUP OF PLAYERS COMPUTED FOR EACH GENERATOR

Grouping Approach	Player Type	Generator R	Generator K	Generator C
Subjectively	Runner	−0.316	**0.211**	0.105
	Killer	0.000	**0.143**	−0.143
	Collector	0.111	−0.250	**0.139**
Objectively	Runner	0.048	−0.190	**0.143**
	Killer	−0.111	**0.079**	0.032
	Collector	−0.154	**0.077**	**0.077**

Bold and underlined texts indicate maximum and minimum, signifying positive and negative correlations, respectively.

3) Alignment Between Personas and Generators: To further test the alignment between player personas and their corresponding generators we calculate the *normalized average rank* for each generator with respect to each player type and present the results in Table III. Particularly, for each generator, we compute its average rank, invert it, and then linearly map it to $[-1, +1]$ so that the higher the value is, the higher the preference for a level. A value of 1 (-1) means that all levels of that category are ranked first (third) among all participants. According to the results obtained, the *Runner* player type appears to prefer K levels and not prefer the R levels as per our analysis earlier. The K and C levels, however, are preferred by their corresponding player personas compared to other generators. For the reassigned persona groups based on in-game behavior, the K levels are still the most preferred by their corresponding player types. The R and K levels however are equally preferred by the *Collector* persona. While *Runner* players still do not seem to prefer *Runner* levels, the preference for K levels switches to C levels.

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TABLE IV
AVERAGE ± STANDARD DEVIATION OF THE NUMBER OF GAPS, LENGTH OF LONGEST GAPS, NUMBER OF ENEMIES, AND NUMBER OF COINS IN GENERATED LEVELS ACROSS EACH OF THE THREE GENERATORS

Generator	#Gaps	Longest Gap	#Enemies	#Coins
R	**2.35±3.22**	**0.71±0.45**	**35.27±6.44**	17.37±5.24
K	0.96±1.39	0.48±0.50	24.09±5.91	**22.35±8.16**
C	1.00±1.48	0.46±0.50	23.37±6.17	22.11±7.76

Significance is evaluated through the wilcoxon signed rank test (the significance level is set to 0:05). Bold and underlined texts indicate maximum and minimum, respectively.

TABLE V
CONFUSION MATRIX OF SELF-REPORTED (SUBJECTIVE) AND BEHAVIORALLY-DEFINED (OBJECTIVE) PERSONAS

		Subjective: Runner	Killer	Collector	N/A	Total #players
Objective	Runner	8	2	8	3	21
	Killer	23	12	23	5	63
	Collector	7	0	5	1	13
Total #players		38	14	36	9	97

Grey cells highlight the number of players who reveal the same persona both subjectively and objectively.

The fact that K and C levels are ranked quite differently by *Killer* and *Collector* types can be surprising as they appear to be similar in terms of both the number of items (cf., Table IV) and f_x values (cf., Table I). However, it is important to note that the similarity in the number of items does not necessarily mean that the levels will elicit similar playing experiences. Players have different preferences and play styles, which can lead to different perceptions of *fun*, even in levels with similar objective characteristics, as discussed in Section VII-B.

To examine the potential inaccuracies in participants' understanding of a persona type when asked through a self-assessment protocol, we directly compare players' *subjectively* reported persona with their *objectively* measured persona types via a confusion matrix (see Table V). The confusion matrix reveals some discrepancies and misalignments between players' self-reported personas and their observed behaviors. For example, many players identified as *Collector*s are behaviorally assessed and categorized as either *Runner*s or *Killer*s. On the other hand, the vast majority of *Killer*s (i.e., 86%) show an alignment between their reported persona and behavioral persona. The inconsistencies between self-reported personas and behavior-identified personas may be due to a combination of 1) various self-reporting and perception biases of each persona type and 2) atypical behaviors of the persona agents. As subjectively defined personas are assigned based on comparisons with agent behaviors, atypical actions exhibited by these agents can affect the accuracy of persona assignments.

VII. DISCUSSION AND OUTLOOK

The results of the user study offer valuable insights into the correlation between the proposed metrics and players' experience (and perception) of *fun* (cf., Experiment-I), as well as the effectiveness of using specifically designed agents to generate game content tailored to specific player types (cf., Experiment-II). Particularly, the deficient results reported raise several key questions that guide future research, focusing on the relationship between metrics and personas. Even though a positive correlation is demonstrated between level-based and gameplay-based *fun* metrics and players' perception of *fun*, as discussed in Section VII-A, there is a need to determine if the metric accurately reflects the nuanced behaviors between different personas, discussed in Section VII-B. Section VII-C highlights the importance of validating the persona modeling process; in other words, how well do these personalized agents mimic human personas? Section VII-D suggests that the *fun* degree of the levels should be revised dynamically, along with players' learning rates. Lastly, we also discovered an asymmetric "Genotype-Phenotype" mapping discussed in Section VII-E.

A. *Correlation Between* Fun *Metrics and Human Assessment*

The analysis of data from Experiment-I demonstrates a positive correlation between both the level-based and gameplay-based *fun* metrics and players' perception of *fun*, validating our hypothesis. Especially, the gameplay-based metric exhibits a stronger correlation compared to the level-based metric, suggesting that dynamic gameplay information plays a crucial role in influencing players' perception of *fun* than static characteristics of level design alone. The combined *fun* metric, as employed in our prior studies [6], shows positive correlation values with player annotations, reinforcing the reliability of our metrics. This indicates that considering multiple facets of game content (gameplay and levels) provides a more comprehensive representation of perceived *fun*.

In this context, the correlation of 0.269 observed in our study is relatively strong compared to values reported in similar research [29], [62]. In another related study, Yannakakis and Hallam [63] report a total agreement correlation coefficient of 0.3888, which, while slightly higher, applies to a different genre and pair of metrics, notably challenge and curiosity; those metrics are cross-verified with reported notions of *fun* in [64]. A correlation of 0.22 is also reported between average response time and self-reported *fun* of children playing physical interactive games [65]. Our study, however, follows Koster's *fun* theory and introduces a set of new metrics related to facets of game content that are tailored to the genre of platformer games. Within platformer games, Pedersen et al. [66] investigated the linear relationships between self-reported emotions and extracted in-game features and the highest correlation value they report (0.345) is attributed to kicking the turtle shell. In contrast, our study attempts to capture the player experience of *fun* holistically through a single ad-hoc designed computational metric rather than a set of extracted gameplay features. Therefore, within our research scope, a correlation of 0.269 may indicate a meaningful link between the computational metrics and the players' perception of *fun*. Furthermore, to our best knowledge, no other work has investigated the correlation between human evaluation of *fun* and computational metrics of *fun* [62], which is worth further investigating in the future.

B. *Multifacet Experience Modeling*

While this study employed play traces as the sole aspect of gameplay for evaluating the gameplay experience, the user

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study results suggest that the f_G metric was only partially effective for constructing personalised generators. Analysis of Experiment-II suggests that personalised EDRL generators still need improvement in matching the preferences of each persona. The data reveals a misalignment between the levels designed for specific player personas (*Runner*, *Killer*, and *Collector*) and the preferences exhibited by players, both through subjective questionnaires and by categorising them based on their in-game behavior. This suggests that our understanding of what defines a player type may not fully align with what players truly enjoy. The observation that similarity in metric values does not necessarily yield similar game experiences (cf., Table IV) is worthy of further investigation.

The gameplay-divergence measurement used in this work evaluates play traces without awareness of specific in-game events, potentially missing subtle differences in gameplay experiences that contribute to varying perceptions of *fun* among different player types. Relying solely on play traces may not sufficiently capture and differentiate the diverse experiences that dissimilar player types might have. Considering and capturing specific in-game events, actions, or other types of metrics [63], [67] in alternative f_G metrics could quantify the gameplay experience in a holistic manner. To enhance the *fun* experience for each persona, future work could investigate the incorporation of gameplay information as part of the input or reward formulation to train personalized generators. This approach would make the generator aware of the player's preferences and behavior during gameplay, allowing for adaptive, personalized content generation tailored to a specific persona.

C. Validation of Persona Modeling

The AI-playing agents used in this study are not designed to mimic human players precisely. For instance, they do not exhibit behaviors, such as stopping to think or going back and forth to collect missed items, which human players may often do. Despite A^* agents simulating different human player personas, there remains a gap between the behavior of the human personas and the corresponding agents. This gap suggests that more nuanced agents are needed to better mimic the complex and varied behaviors of real players. Future work should focus on incorporating a broader range of human-like behaviors and human-like agents [68].

D. Player Engagement via Dynamic Experience Modeling

According to Koster's *theory of fun*, a game is considered *fun* when players are constantly learning new skills and improving during gameplay [6]. However, our work does not account for a player's learning ability during the level generation process. The metrics used may not fully align with a player's temporal perception of *fun*. In addition, assumptions about player personas and their consistency over time may not hold, as indicated by their performance. Future work could incorporate skill conditioning [69] into both metrics and gameplay agents to directly consider a player's skills and learning ability during level generation. This approach could potentially create more tailored and enjoyable gaming experiences and also increase the responsiveness of generators, ensuring that they can adapt over time to reflect changes in player strategies and preferences.

E. Asymmetric "Genotype-Phenotype" Mapping

The collection of test levels for Experiment-I (cf., Fig. 4) indicates that levels with very high or very low metric values are under-represented even though all levels were obtained by sampling latent vectors uniformly at random. This observation implies an asymmetric mapping between "genotype" (i.e., latent vector) and "phenotype" (i.e., f_L and f_G of decoded level). A potential future work is to investigate the relationship between latent vectors and the *fun* value of levels, and further analyse the playing experience landscape.

VIII. Conclusion

In this work, we presented the results of an extensive user study, involving more than 90 participants, designed for the purpose of evaluating the effectiveness of our experience-driven PCG via reinforcement learning framework [2], [6]. Particularly, the study has a twofold aim and corresponding research hypotheses. First, we attempt to cross-verify that the *fun* metrics introduced as reward functions for EDRL [2], [6] are correlated with human notions of *fun* in this game. Second, we tested the degree to which our EDRL generators can create personalised levels while considering individual persona preferences. Obtained results reveal that the combined *fun* metric $f_L + f_G$ for the two creative facets of gameplay and level [7] yields significant correlation values with human notions of *fun* cross-verifying our proposed metric and validating our first hypothesis. Findings also suggest that a comprehensive evaluation of *fun* that considers more than one creative facet of game design is better aligned with the human notion of *fun* as expressed via preferences. The individual *fun* metrics for each one of the two facets are also positively correlated with players' perception of *fun*. Compared to the level-based metric (f_L), however, the gameplay-based metric (f_G) yields higher (and significant) correlation values with human preferences. Overall, our findings suggest that quantifiable notions of *fun* that attempt to capture moderate levels of game content divergence are correlated with human notions of perceived *fun*.

Our player-adaptive generation experiments—using three specifically designed agents to generate game content tailored to corresponding human player types—reveal promising yet inconclusive results. While it appears that experience-driven and personalised game adaptation is possible, the degree to which this is successful varies across players and player types.

The EDRL framework is directly applicable to any game content generator (beyond platformers) as long as an efficient and compact content representation is available and experience metrics are sufficiently reliable. Further research in the directions mentioned in Section VII could lead to more comprehensive PEM, and more efficient and accurate content generation methods, enabling game developers to create personalised and engaging experiences for players of various types and preferences. By bridging the gap between player behavior and content

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generation, future PCG models can better cater to the personalized needs and desires of players, ultimately enhancing the overall gaming experience.

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Ziqi Wang (Member, IEEE) received the B.E. degree in computer science and M.E. degree in electronics science from the Southern University of Science and Technology, Shenzhen, China, in 2021 and 2024, respectively.

His research focuses on concentrates on game content generation through artificial intelligence methods including reinforcement learning, generative adversarial networks, and evolutionary computation.



Yuchen Li (Student Member, IEEE) received the B.S. degree in integrated digital media and the M.A. degree in interactive media arts in 2021 and 2022, respectively, from New York University (NYU), New York, NY, USA, where she is currently working toward the Ph.D. degree in computer science under the supervision of Prof. Julian Togelius.

Her research focuses on procedural content generation (PCG), particularly human-centered PCG and the application of large language models in PCG.



Haocheng Du (Student Member, IEEE) received the B.E. degree in computer science from the Southern University of Science and Technology (SUSTech), Shenzhen, China, in 2023. He is currently a postgraduate student at Tsinghua Shenzhen International Graduate School.

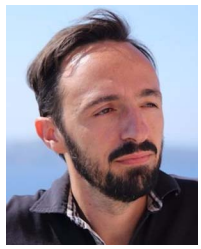
His current research focuses on game design, with a primary focus on game design theory, and reflective game design and development.



Jialin Liu (Senior Member, IEEE) received the B.E. degree in optoelectronic information science and engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2010, the Diplôme d'Ingénieur in computer science from the Polytech'Paris-Sud, Gif-sur-Yvette, France, in 2012, the M.Sc. degree in bioinformatics and biostatistics from the École Polytechnique and the Université Paris-Sud in 2013, and the Ph.D. degree in computer science from the Université Paris-Saclay in 2016.

Currently, she is a Tenure-Track Assistant Professor with the Department of Computer Science and Engineering, Southern University of Science and Technology (SUSTech), Shenzhen, China. Before joining SUSTech, she worked with the University of Essex, Colchester, U.K., as a Senior Research Officer and Queen Mary University of London, London, U.K., as a Postdoctoral Research Associate. Her main research interests include AI in games, evolutionary computation, fair machine learning, optimization and learning in uncertain environments.

Dr. Liu is an Associate Editor for IEEE TRANSACTIONS ON GAMES, IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION, IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE, and the *Knowledge-Based Systems*.



Georgios N. Yannakakis (Fellow, IEEE) received the Ph.D. degree in informatics from the University of Edinburgh, Edinburgh, Scotland, in 2006.

He is a Professor with the Institute of Digital Games, University of Malta (UM), and a co-founder of modl.ai, Copenhagen, Denmark. He is the coauthor of the *Artificial Intelligence and Games textbook* and the co-organiser of the Artificial Intelligence and Games summer school series. He has authored or coauthored more than 350 papers in the area of his research fields, which includes crossroads of artificial intelligence, affective computing, games, and computational creativity.

Dr. Yannakakis was the recipient of numerous national and European grants, including a Marie Skåodowska-Curie Fellowship, and IEEE TRANSACTIONS ON GAMES Outstanding Paper Award for his several papers. He has appeared in *Science Magazine* and New Scientist among other venues. He is currently the Editor-in-Chief of IEEE TRANSACTIONS ON GAMES, an Associate Editor for IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION, and he was an Associate Editor for IEEE TRANSACTIONS ON AFFECTIVE COMPUTING, IEEE TRANSACTIONS ON COMPUTATIONAL INTELLIGENCE, and *AI in Games* journals.

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